Electronic Devices & Circuits Laboratory Experiment 4: NMOS Common-Source Amplifier with Resistive Load: DC Transfer Characteristic and Small-Signal Gain

Bhargav K (EE25BTECH11013)
Department of Electrical Engineering
IIT Hyderabad

Aditya Appana (EE25BTECH11004)
Department of Electrical Engineering
IIT Hyderabad

September 4, 2026

Contents

1 Objective **3**

2 Circuit **3**

3 Finding the threshold of the NMOS **3**

3.0.1 Extraction of Threshold Voltage 4

4 Part A - DC Transfer Characteristic **6**

4.1 Procedure 6

4.2 Simulation 7

4.3 Observations 7

5 Part B - Small-Signal Gain vs. DC Bias Point (Two Methods) **9**

5.1 Method 1: Slope of the measured transfer characteristic 9

5.2 Method 2: Direct AC measurement 10

5.3 Simulation 10

5.3.1 Small-Signal AC Measurement at $V_{IN,DC} = 1.5$ V 12

6 Analysis and Discussion Questions **13**

7 Precautions **16**

1 Objective

To construct a common-source (CS) NMOS amplifier with a resistive drain load, measure and plot its DC transfer characteristic (V_{OUT} vs V_{IN}), extract the small-signal voltage gain as a function of the DC bias point in two independent ways, and verify that the two agree.

2 Circuit

The gate of an enhancement NMOS is driven by V_{IN}. The drain is connected to V_{DD} through a resistor R_D and is the output node, V_{OUT}. The source is grounded. A DC supply sets V_{IN} through a potentiometer or a programmable source; a small AC signal from a function generator is superposed on V_{IN} through a coupling capacitor and blocking resistor for Part B.

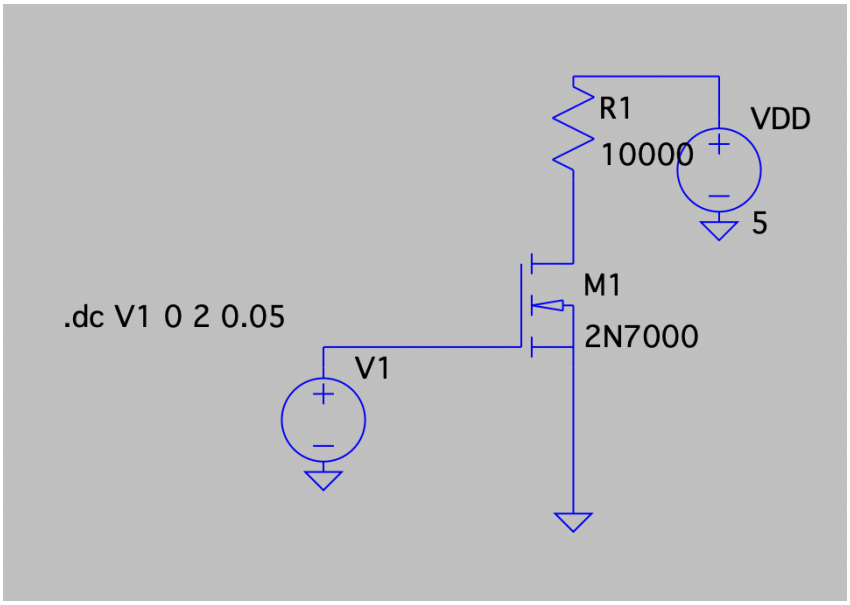


Figure 1: Circuit

3 Finding the threshold of the NMOS

This is already done in the previous experiment, but we are doing it again because we are using a new MOSFET and it will have different properties(like threshold voltage). Before analyzing the common-source amplifier, the NMOS device was characterized to determine important device parameters. The transfer characteristic I_D versus V_{GS} was measured while maintaining the drain voltage at approximately 5 V. The measured data are shown in Table 1.

Table 1: I_D versus V_{GS} characteristics of the NMOS

V_{GS} (V)	I_D (mA)
0.00	0
0.10	0
0.20	0
0.30	0
0.40	0
0.50	0
0.60	0
0.70	0
0.80	0
0.90	0
1.00	0
1.05	0
1.10	0
1.15	0.0003
1.20	0.0005
1.25	0.0011
1.30	0.0023
1.32	0.0033
1.34	0.0043
1.36	0.0062
1.38	0.0085
1.40	0.0118
1.42	0.0163
1.44	0.0220
1.46	0.0300
1.48	0.0407
1.50	0.0548
1.52	0.0736
1.54	0.0981
1.56	0.1300
1.58	0.1700
1.60	0.2223
1.62	0.2877
1.64	0.3680
1.66	0.4708
1.68	0.5964
1.70	0.7502

3.0.1 Extraction of Threshold Voltage

For an NMOS operating in saturation and following the square-law model,

$$I_D = \frac{k_n}{2}(V_{GS} - V_{TH})^2. \tag{1}$$

For finding V_{TH} :

$$I_D = \frac{k_n}{2}(V_{GS} - V_{TH})^2. \tag{2}$$

Taking the square root,

$$\sqrt{I_D} = \sqrt{\frac{k_n}{2}}(V_{GS} - V_{TH}). \tag{3}$$

Therefore, a plot of $\sqrt{I_D}$ versus V_{GS} should be approximately linear in the appropriate saturation region. The linear equation can be written as

$$\sqrt{I_D} = mV_{GS} + c, \tag{4}$$

where m is the slope and c is the intercept.
At the threshold voltage,

$$I_D = 0, \tag{5}$$

and hence

$$0 = mV_{TH} + c. \tag{6}$$

Therefore,

$$\boxed{V_{TH} = -\frac{c}{m}} \tag{7}$$

For the measured data, the approximately linear region from $V_{GS} = 1.50$ V to 1.70 V was used for the linear fit. The fitted relation was found to be approximately

$$\sqrt{I_D} = 3.136\, V_{GS} - 4.515, \tag{8}$$

when I_D is expressed in amperes and $\sqrt{I_D}$ is expressed in $\sqrt{\text{A}}$.
Hence,

$$V_{TH} = -\frac{-4.515}{3.136} \approx 1.440 \text{ V}. \tag{9}$$

Thus, the threshold voltage extracted from the measured transfer characteristic is

$$\boxed{V_{TH} \approx 1.44 \text{ V}}. \tag{10}$$

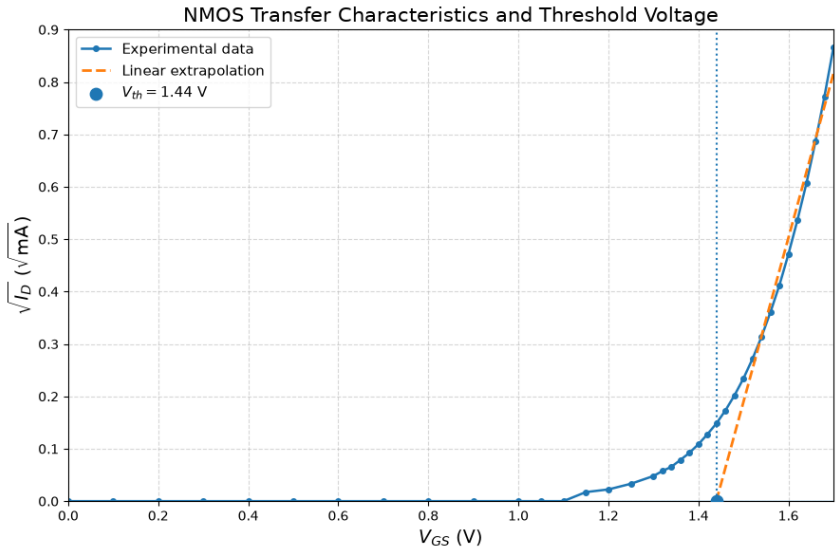


Figure 2: Threshold calculations

The first observable non-zero drain current occurs at a voltage slightly above 1.1 V, but this point is not taken as the threshold voltage because practical measurements contain finite resolution and leakage currents. Instead, the threshold voltage is obtained from the extrapolated intercept of the approximately linear $\sqrt{I_D}$ versus V_{GS} characteristic.

4 Part A - DC Transfer Characteristic

4.1 Procedure

- Sweep V_{IN} from 0 V to V_{DD} in steps of about 0.2 V (finer steps of 0.05 V near your device's V_{TH} and near the saturation-to-triode boundary).
- At each step, record V_{OUT} with the oscilloscope. Do not let I_D exceed your device's datasheet current limit.
- The three regions on the plot: cutoff, saturation (the amplifying region), and triode to be plotted.
- Plot V_{OUT} vs V_{IN} on linear axes for the full sweep.

4.2 Simulation

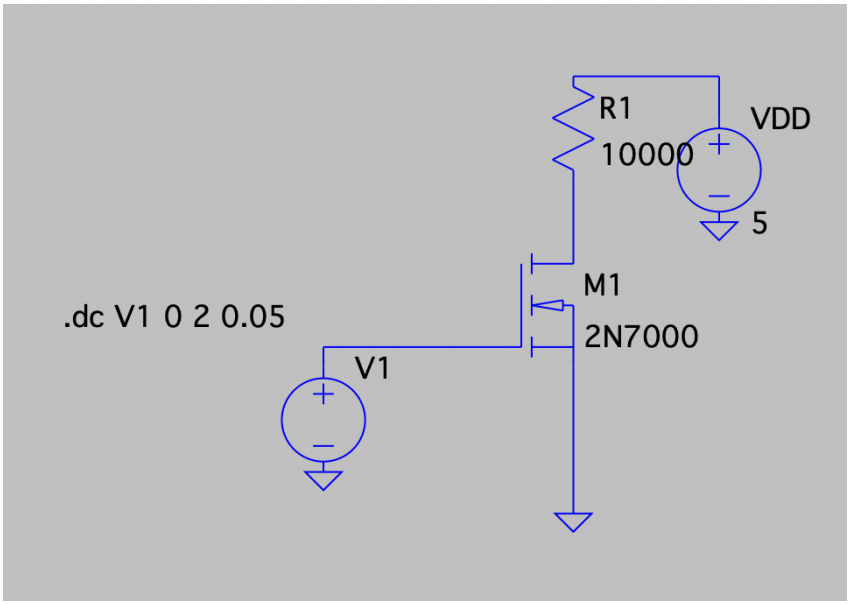


Figure 3: Circuit

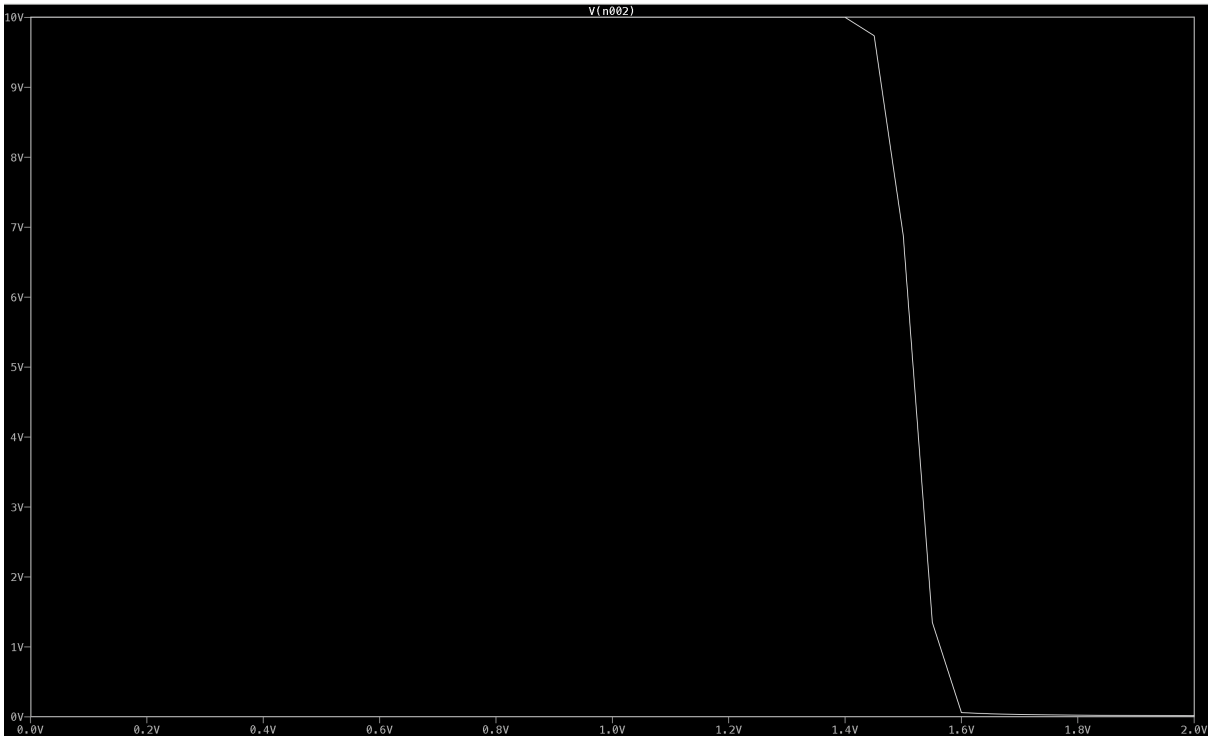


Figure 4: LTSpice simulation

4.3 Observations

The following table summarizes the measured data for the DC transfer characteristics of the NMOS transistor.

Table 2: DC Transfer Characteristic Data (Threshold = 1.43V, VDD = 5V)

No.	V_{GS} **or** V_{in} **(V)**	V_{DS} **(V)**	**Region**
1	0	4.9984	Cutoff
2	0.1	4.9984	Cutoff
3	0.2	4.9984	Cutoff
4	0.3	4.9984	Cutoff
5	0.4	4.9984	Cutoff
6	0.5	4.9984	Cutoff
7	0.6	4.9984	Cutoff
8	0.7	4.9984	Cutoff
9	0.8	4.9984	Cutoff
10	0.9	4.9984	Cutoff
11	1	4.9984	Cutoff
12	1.05	4.9983	Cutoff
13	1.1	4.9982	Cutoff
15	1.2	4.9968	Cutoff
17	1.3	4.99	Cutoff
19	1.4	4.9578	Cutoff
20	1.42	4.943	Cutoff
21	1.44	4.923	Amplifier
22	1.46	4.8964	Amplifier
23	1.48	4.8607	Amplifier
24	1.5	4.8128	Amplifier
25	1.52	4.7503	Amplifier
26	1.54	4.6684	Amplifier
27	1.56	4.562	Amplifier
28	1.58	4.425	Amplifier
29	1.6	4.251	Amplifier
30	1.62	4.03	Amplifier
31	1.64	3.764	Amplifier
32	1.66	3.425	Amplifier
33	1.68	3.014	Amplifier
34	1.7	2.507	Amplifier
35	1.72	1.9	Amplifier
36	1.74	1.181	Amplifier
37	1.76	0.391	Amplifier
38	1.78	0.139	Triode
39	1.8	0.091	Triode
40	1.9	0.034	Triode
41	1.95	0.026	Triode

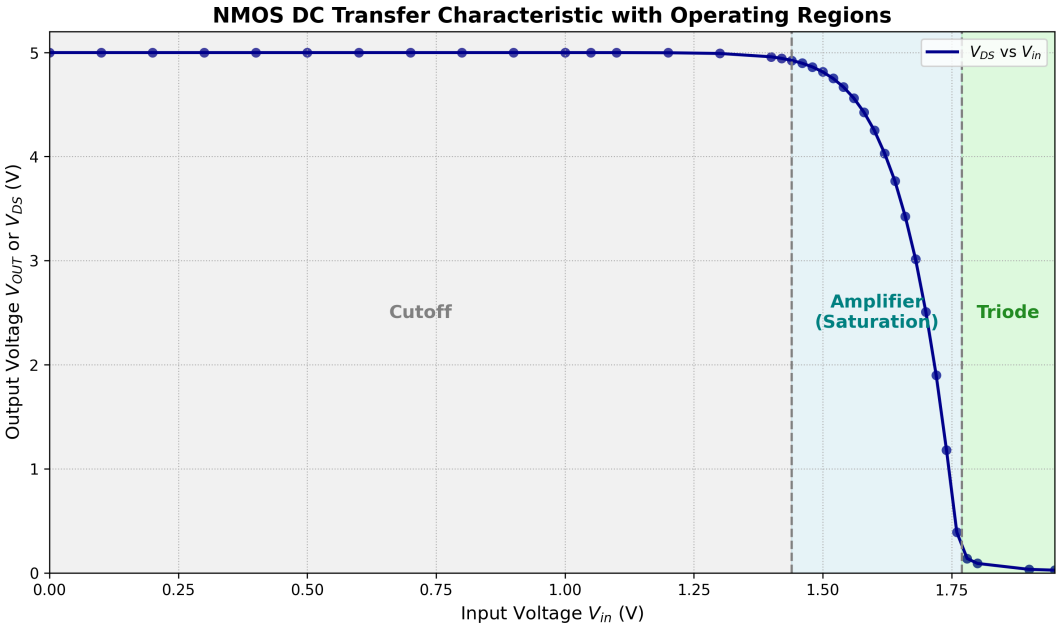


Figure 5: Graph showing the cutoff, amplifier, and triode regions

5 Part B - Small-Signal Gain vs. DC Bias Point (Two Methods)

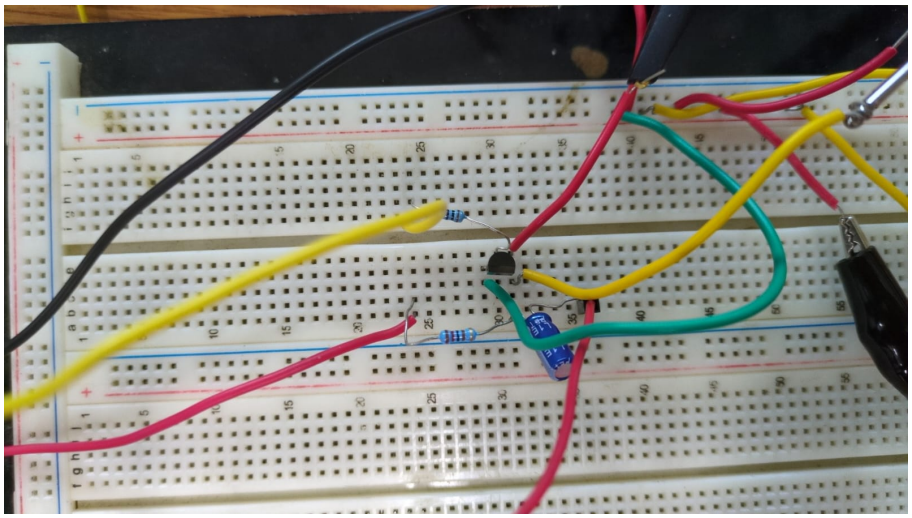


Figure 6: Circuit Diagram

5.1 Method 1: Slope of the measured transfer characteristic

Compute $A_v(V_{IN}) = d(V_{OUT})/d(V_{IN})$ as the numerical derivative (point-to-point slope) of the V_{OUT} vs V_{IN} data from Part A. This does not require any additional measurement.

5.2 Method 2: Direct AC measurement

5.3 Simulation

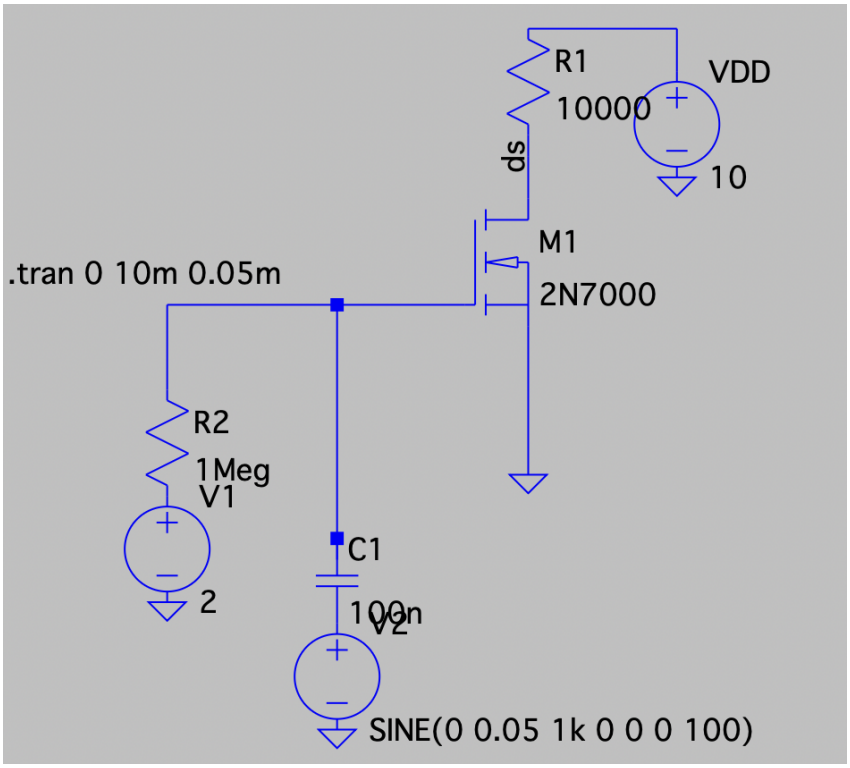


Figure 7: Circuit

For a chosen set of DC bias points $V_{IN,DC}$ (span cutoff edge, mid-saturation, and near-triode for the device), superpose a small AC tone (1 kHz) on V_{IN}. On the oscilloscope, measure the AC output amplitude and divide by the AC input amplitude: $A_v = -v_{out,ac,pp}/v_{in,ac,pp}$.

The small-signal analysis was performed to determine the voltage gain of the NMOS common-source amplifier at different DC bias points. A small sinusoidal AC signal was superimposed on the DC gate voltage, and the resulting output voltage was measured using a digital storage oscilloscope (DSO). The gain was then determined from the ratio of the output and input AC amplitudes.

The total input voltage applied to the gate can be represented as

$$V_{IN}(t) = V_{IN,DC} + v_{in}(t), \tag{11}$$

where $V_{IN,DC}$ is the selected DC bias voltage and $v_{in}(t)$ is the small AC signal. The AC input used in the experiment had an amplitude of approximately 10 mV at a frequency of approximately 1 kHz.

For a common-source NMOS amplifier, an increase in the gate-to-source voltage increases the drain current. This increases the voltage drop across the drain resistor and consequently decreases the drain voltage. Therefore, the output signal is inverted with respect to the input signal, giving a negative voltage gain.

The small-signal voltage gain is defined as

$$A_v = \frac{v_{out}}{v_{in}}. \tag{12}$$

Experimentally, the gain was obtained from the ratio of the measured output and input signal amplitudes. Since the common-source amplifier introduces a phase inversion, the gain is written as

$$\boxed{A_v = -\frac{V_{out,AC}}{V_{in,AC}}} \tag{13}$$

For a sufficiently small sinusoidal signal, the output remains approximately sinusoidal and the above ratio represents the small-signal voltage gain.

The small-signal gain was measured for several values of the DC gate voltage. The measured results are shown in Table 3.

Table 3: Measured small-signal AC gain at different DC bias points

$V_{IN,DC}$ (V)	V_{RMS} (mV)	A_v
0.00	62.0	-8.7668
0.20	62.0	-8.7668
0.40	62.0	-8.7668
0.60	62.0	-8.7668
0.80	62.0	-8.7668
1.00	62.0	-8.7668
1.20	62.0	-8.7668
1.30	85.0	-12.0190
1.35	150.0	-21.2100
1.40	310.0	-43.8340
1.45	640.0	-90.4960
1.50	7.2	-1.0181
1.55	2.0	-0.2828
1.60	0.83	-0.1174
1.65	0.40	-0.0566
1.70	0.23	-0.0325
1.75	0.00	0
1.80	0.00	0

At $V_{IN,DC} = 1.50$ V, the measured AC voltage gain is

$$A_v = -1.0181. \tag{14}$$

For a common-source amplifier,

$$A_v \approx -g_m R_D. \tag{15}$$

Hence,

$$g_m = \frac{|A_v|}{R_D} = \frac{1.0181}{2.2 \times 10^3} = 4.628 \times 10^{-4}\,\text{S}. \tag{16}$$

Therefore,

$$\boxed{g_m \approx 0.463 \text{ mS}} \tag{17}$$

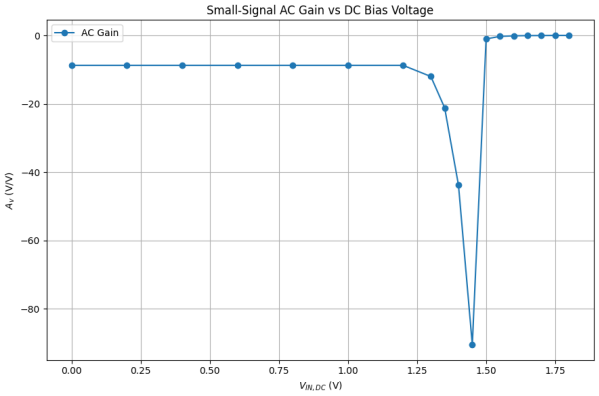


Figure 8: Small signal gain vs DC operating point

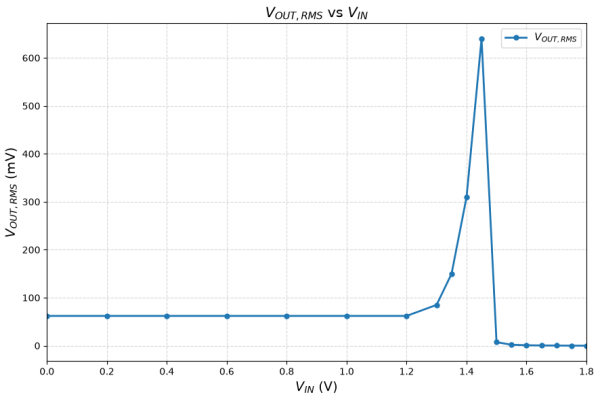


Figure 9: RMS graph

5.3.1 Small-Signal AC Measurement at $V_{IN,DC} = 1.5$ V

A small-signal AC analysis was performed by biasing the NMOS common-source amplifier at a DC gate voltage of

$$V_{IN,DC} = 1.5 \text{ V} \tag{18}$$

and superimposing a sinusoidal input signal with an amplitude of

$$V_{in,AC} = 50 \text{ mV}. \tag{19}$$

The corresponding output waveform was observed using the digital storage oscilloscope (DSO). The output waveform was found to be a clean sinusoidal waveform without significant distortion or visible clipping. This indicates that the amplifier was operating satisfactorily around the selected DC bias point.

The AC voltage gain was determined from the ratio of the output and input AC amplitudes:

$$A_v^{AC} = -\frac{V_{out,AC}}{V_{in,AC}} = 8.925. \tag{20}$$

The negative sign represents the phase inversion inherent to the common-source amplifier.

The experimentally measured AC gain at $V_{IN,DC} = 1.5$ V was found to be approximately equal to the DC gain obtained from the slope of the DC transfer characteristic,

$$A_v^{DC} = \frac{dV_{OUT}}{dV_{IN}} = 9.6. \tag{21}$$

Thus,

$$\boxed{A_v^{AC} \approx A_v^{DC}} \tag{22}$$

at the selected operating point. The close agreement between the two gain values, together with the clean sinusoidal output waveform, confirms that the amplifier exhibits the expected small-signal behavior around the 1.48 V DC bias point.

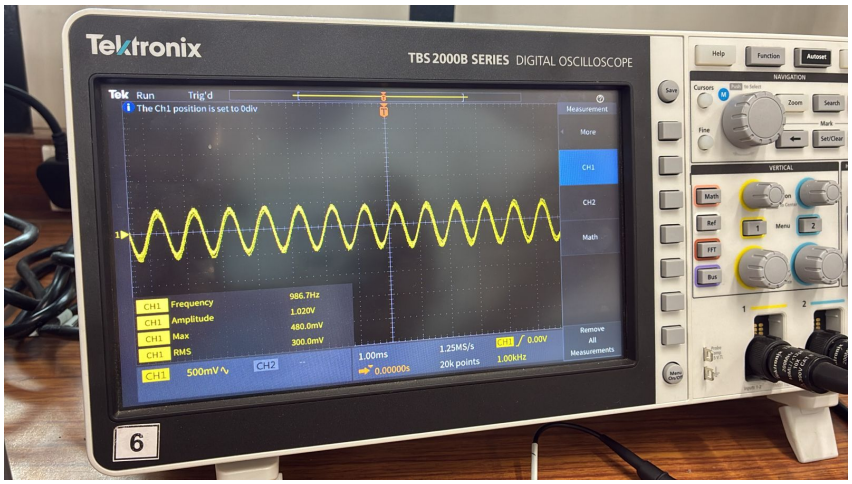


Figure 10: 1.48V DC Gate Voltage

6 Analysis and Discussion Questions

1. **Where along the V_{IN} axis is the gain magnitude largest for your device? Relate this to g_m and to how close V_{OUT} sits to the edge of saturation.**

   From the measured DC transfer characteristic, the magnitude of the voltage gain increases significantly as V_{IN} is increased from the threshold region and reaches its maximum near the saturation-to-triode transition, at approximately

   $$V_{IN} \approx 1.70\text{–}1.72 \text{ V}. \tag{23}$$

   In the saturation region, the small-signal voltage gain is approximately

   $$A_v \approx -g_m R_D. \tag{24}$$

As V_{IN} increases, the transconductance g_m increases, resulting in an increase in the gain magnitude. The gain becomes largest when the transistor has a relatively large g_m while V_{OUT} is approaching the edge of saturation. Beyond this point, the transistor enters the triode region and the gain magnitude decreases.

2. **Why does the gain fall toward zero in cutoff and again deep in triode?**

   In the cutoff region, the NMOS is essentially OFF and

$$I_D \approx 0, \tag{25}$$

   so a change in V_{GS} produces very little change in drain current. Therefore,

$$g_m \approx 0 \tag{26}$$

   and consequently the voltage gain approaches zero.

   In the deep triode region, the drain voltage becomes very small and the transistor no longer behaves as a high-output-resistance current source. The output voltage becomes much less sensitive to small changes in V_{GS}, causing the magnitude of the voltage gain to decrease again toward zero.

3. **How closely do your Method 1 and Method 2 gain values agree at each bias point? Account for any discrepancy.**

   Method 1 determines the gain from the slope of the measured DC transfer characteristic,

$$A_v^{(1)} \approx \frac{\Delta V_{OUT}}{\Delta V_{IN}}, \tag{27}$$

   whereas Method 2 determines the gain directly from the measured AC amplitudes,

$$A_v^{(2)} = -\frac{V_{out,AC}}{V_{in,AC}}. \tag{28}$$

   The two methods showed good agreement at the bias points where the transistor was operating sufficiently close to the small-signal condition. For example, at

$$V_{IN,DC} = 1.5 \text{ V}, \tag{29}$$

   the measured DC and AC gain values were approximately equal. This agreement confirms the validity of the small-signal approximation at that operating point.

   Some discrepancy was observed at other bias points, particularly near the cutoff and transition regions. Possible causes include measurement noise, the finite step size of the DC sweep, uncertainty in determining the local slope. These effects become more significant when the transistor characteristic changes rapidly with V_{IN}.

4. **If R_D is increased, how would you expect the saturation-region gain and the extent of the triode region to change?**

   In the saturation region, the voltage gain is approximately

$$A_v \approx -g_m R_D. \tag{30}$$

   Therefore, increasing R_D increases the magnitude of the voltage gain, provided that the transistor remains in saturation.

   However, increasing R_D also produces a larger voltage drop across the resistor for a given drain current,

$$V_{OUT} = V_{DD} - I_D R_D. \tag{31}$$

   Hence, V_{OUT} reaches lower values for the same V_{IN}, causing the transistor to enter the triode region at a lower value of V_{IN}. Therefore, the triode region extends over a larger portion of the input-voltage range, while the saturation-region range becomes smaller.

5. **Why does the slope of the large-signal DC transfer curve at a bias point equal the small-signal AC gain measured at that same bias point?**

   For a sufficiently small variation around a DC operating point,

$$\Delta V_{OUT} \approx \frac{dV_{OUT}}{dV_{IN}} \Delta V_{IN}. \tag{32}$$

   Therefore,

$$\boxed{A_v^{AC} \approx \frac{dV_{OUT}}{dV_{IN}} = A_v^{DC}} \tag{33}$$

   Thus, the small-signal AC gain is simply the local slope of the large-signal DC transfer characteristic at the chosen operating point.

6. **Compare your device's V_{TH} and the V_{IN} at which your gain peaks against the illustrative Section 6 values.**

   From the measured transfer characteristic, the threshold voltage of the device was extracted as approximately

$$V_{TH,\text{measured}} \approx 1.44 \text{ V}. \tag{34}$$

   The peak gain is achieved when :

$$V_{TH,\text{i}} = 1.45 \text{ V}. \tag{35}$$

   Therefore, the measured threshold voltage is lower by approximately

$$\Delta V_{TH} = 1.44 - 1.45 = -0.01 \text{ V}. \tag{36}$$

Hence, the device's threshold voltage is approximately

$$0.01 \text{ V lower than the illustrative value.} \tag{37}$$

From the measured DC transfer characteristic, the magnitude of the gain reaches its maximum at approximately

$$V_{IN,\text{peak}} \approx 1.48\text{V}. \tag{38}$$

7 Precautions

- It is very important not to increase the V_{GS} more than 2.2V when a peak current flow is observed in our case.
- Keep the AC test signal amplitude small (a few mV) so the small-signal approximation remains valid.
- It is very important to use a coupling capacitor to let the AC signal pass without disturbing the DC input bias
- For this experiment, it is crucial that a DC bias is used so that it first crosses the Threshold value of voltage and be present in the saturation region(amplification) and then does magnification, instead of just passing only the input AC Signal where the input voltage doesnt cross the threshold voltage which is the bare minimum requirement for the flow of current.
- For the particular NMOS, it is crucial to monitor the current flow and always ensure that the current flowing doesn't exceed the maximum current which can flow through the MOSFET (around 2mA) which is present in the datasheet.
- We also faced a problem while taking the values for the I_d vs V_{gs} graph. When I kept the $V_{gs} = 0V$ and $V_{ds} = 1.2V$, I saw a current flow(which is not supposed to happen). Then I realised that the MOSFET had an issue in it and changed the MOSFET and then it started working.
- It is important to ensure that the MOSFET is not connected to the power source for a long time because it will heat up and it will make the MOSFET non-working.
- Before switching to a different circuit and before switching on the power supply, always ensure the input voltage starts from 0V otherwise there might be issue in heating up or wrong values for measurements. Because from the previous experiment which might have required higher input voltage might directly affect the circuit and cause high heat.
- If there is a lot of noise in the DSO for the AC small-signal analysis in the output, then ensure that a resistor ladder for the input AC is used so that it finally gives a required input sin wave which is of a required amplitude, since the the noise is also diminshed when a resistor ladder is used.